Eli Doggart

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Experience

Nimbus Power Systems
Mechanical Design and Manufacturing Engineer **May 2022–Present**

- Designed, optimized, fabricated, and scaled up to prototype a four-fluid hydrogen fuel cell implementing performance, efficiency, and cost saving improvements (see projects for detail).
- Designed, built, wired, and plumbed 3 test stands for small-scale and engine-scale fuel cell testing, addressing safety for 1,000-volt wiring and flammable pressurized gases.
- Designed and built a facility-wide hydrogen safety system with emergency stops and sensors that trigger gas shutoffs, alarms, and facility exhaust fans.

Accelerated design LLC | *Design Engineering Intern* **January–March 2022**

- Revised CAD models, created drawings, operated SLA 3D printers.

General Dynamics Electric Boat | *Engineering Intern* **June–August 2021**

- Reviewed supplier requests to deviate from design for the pumps department for Columbia class.

Mott Corporation | *R&D Co-op* **January–June 2021**

- Ran testing equipment, improved lab procedures, conducted studies with metal DMLS printer tuning it for intentional porosity improving repeatability.

East Coast Precision Manufacturing | *Machine operator* **December 2020–January 2021**

- Operated various CNC lathes and mills to produce polymer parts.

Nomad Metallurgy | *Metallurgical Lab Assistant* **June 2016–August 2020**

- Metallurgical cut/mount/polish, microscopy, and hardness testing.

Niantic Bay Bicycles | *Bike Mechanic* **May 2016–August 2020**

- fixed and sold bikes, maintained stock, assisted with office tasks.

Selected Projects

Fuel cell design and fabrication | Nimbus Power Systems **2023-2026**
Designed, simulated, and optimized a 4 fluid hydrogen fuel cell for automotive and aerospace applications. Evenly distributed flow in two dimensions across the active area is extremely important for performance. Constrained by manufacturing limitations and legacy tooling, I reduced the maldistribution in reactants from over 40% to less than 2%. Properly loading flow fields in the active area of the hydrogen fuel cell leads to lower electrical resistance, but too much pressure can restrict fluid flow. I used flow and structural simulation to ensure acceptable levels of pressure and electrical resistance despite a reduction in landing area. After designing this fuel cell, I worked with component suppliers across the globe as well as scaled up in house procedures and facilities in order to produce an engine sized prototype.

Capstone Project | WPI **March–December 2022**
Worked as designer and fabricator on team to produce, and optimize an active suspension system for a longboard.

Camper Van Build | Home **2025-Present**
Designed and fabricated a camper build for a Ram ProMaster including an off grid electrical system, plumbing, innovative bed and floor design. It is modular and easily adaptable due to use of extruded aluminum profiles.

Technical Skills

Computer: SOLIDWORKS (CAD, Flow Simulation, CAM), Fusion (CAD CAM), LabVIEW, LaTeX
Physical: GD&T, CNC and manual lathe and mill, 3D printing (FDM, SLA, DMLS), Basic MIG/TIG welding, wiring
Certifications: Six Sigma Black Belt, Trimech SOLIDWORKS Flow, Summit Toyotalift Forklift

Education

Worcester Polytechnic Institute (WPI), Worcester, MA **May 2023**
Bachelor of Science in Mechanical Engineering, GPA 3.63